

Cluster-Based Robotic Cotton Harvesting For Efficient Cotton Picking

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Introduction

- Automation and robotics are increasingly being adopted in agriculture, including cotton harvesting.
- Robotic cotton harvesting systems offer autonomous operation, scalability, multi-pass harvesting with the potential to reduce fiber quality degradation and yield loss, and improved sustainability; however, no commercially viable system currently exists due to insufficient harvesting speed and efficiency.
- Existing robotic cotton harvesting studies largely follow single-boll detection and picking paradigm, suffering slow operational speed and limited field-scale practicality.
- Vision-guided single-boll harvesting is time- and resource-inefficient for cotton because of high boll density and low individual boll value.

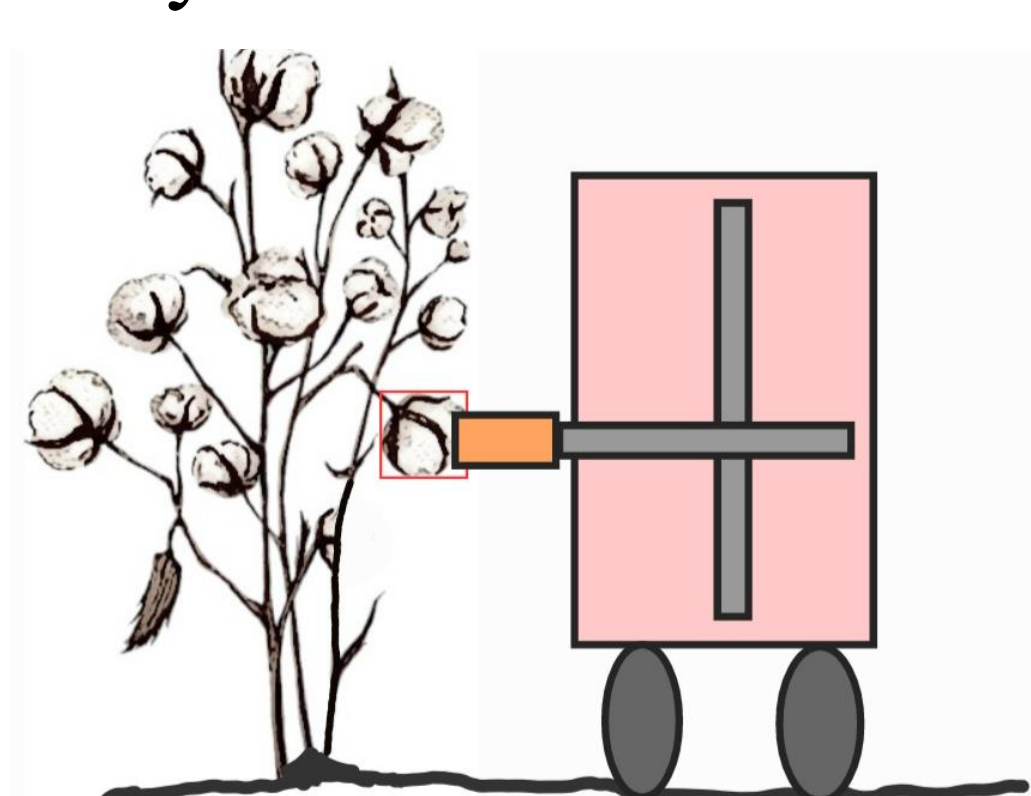


Fig. 1: Harvesting single cotton boll at a time approach, slow

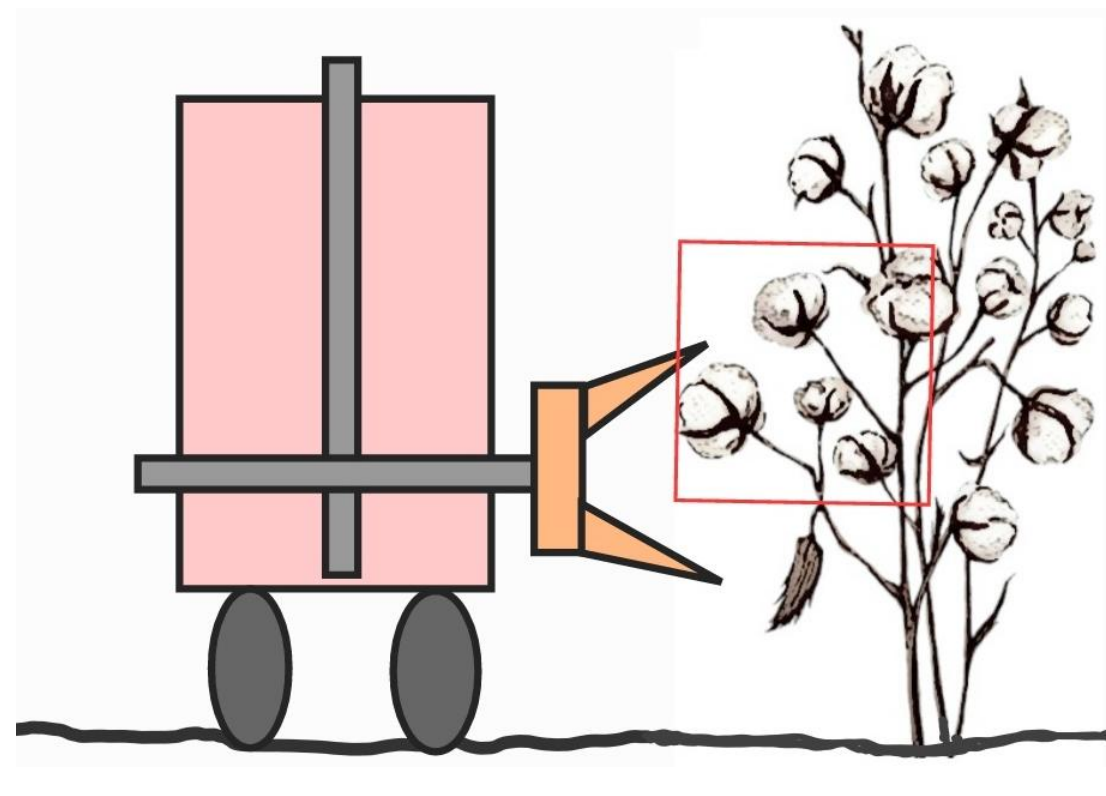


Fig. 2: Harvesting multiple cotton bolls at a time approach, faster

- We propose a vision-guided cotton cluster harvesting approach with a novel cotton picker end-effector designed to improve picking efficiency, access inner canopy regions, and reduce trash collection.

Objective

The primary goal of this study was to evaluate the performance of a barbed-spindle end-effector for robotic harvesting of cotton boll clusters. The specific objectives were to:

- Design and fabricate a cotton picker spindle array (CPSA) end-effector.
- Determine the optimal picking approach, spindle speed, and overall system efficiency.

Materials and Methods

I. End-Effector Versions

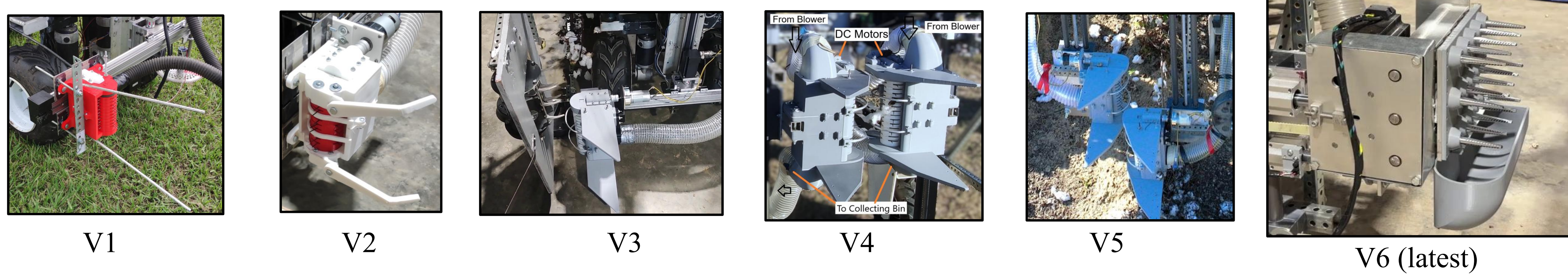


Fig. 3: Previously studied end-effector versions (V1–V5) based on a roller-tine cotton boll stripping mechanism, which struggled to harvest cotton from the inner canopy. The latest version (V6) uses a barbed spindle array, enabling effective cotton picking from both the inner and outer regions of the canopy.

II. Cotton Picker Spindle Array (CPSA) End-Effector Design and Doffing Mechanism

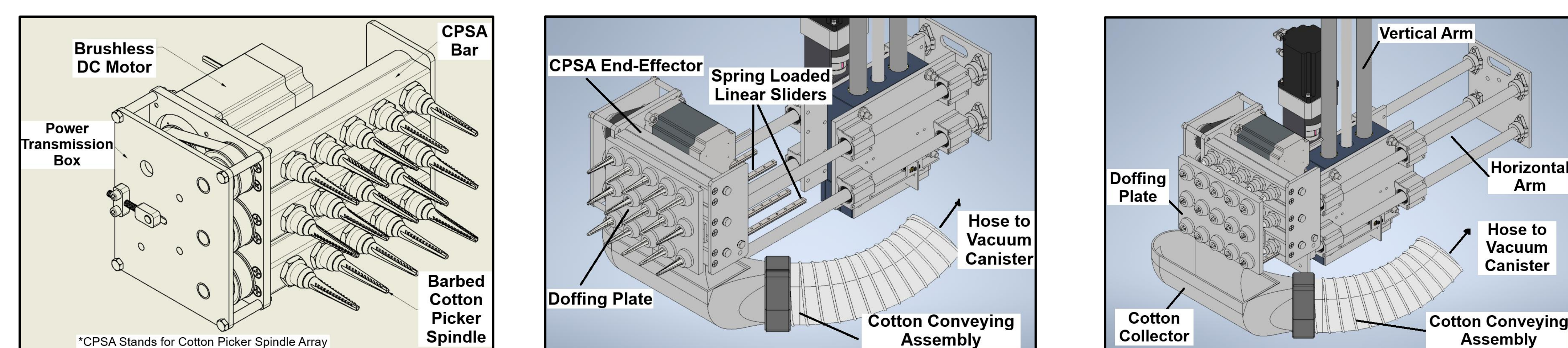
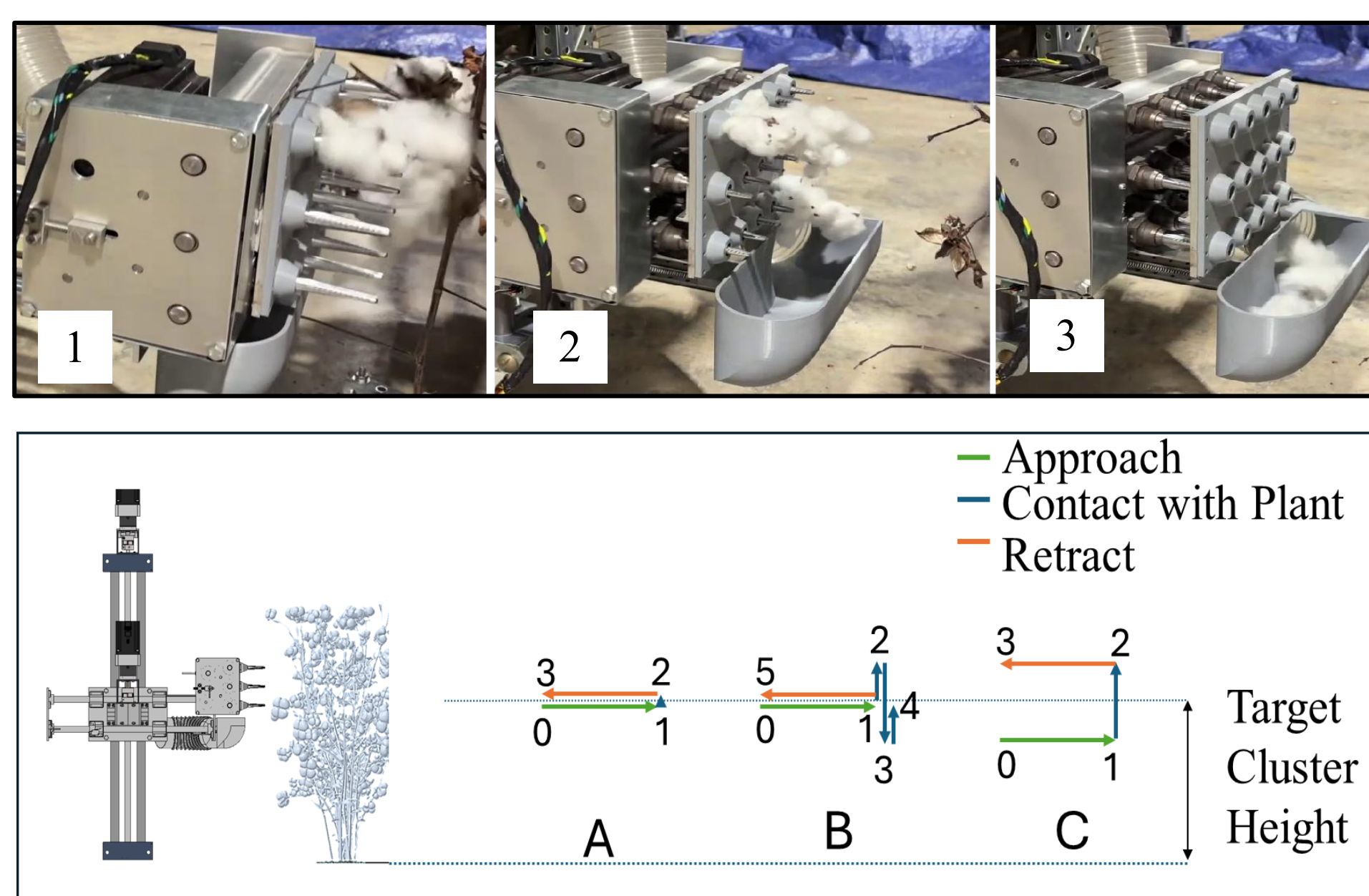


Fig. 4: CPSA end-effector CAD design (left) and the doffing mechanism with its subcomponents (middle and right). The end-effector consists of three cotton picker spindle bar assemblies arranged in a rectangular array, forming a total of 15 barbed picker spindles barbed picker spindles.

III. Picking Approach/ End-Effector Trajectory

- The end-effector approaches the target cotton cluster, and the spindles begin rotating.
- The end-effector retracts, and spindle rotation stops midway through retraction.
- Upon full retraction, the doffing plate removes the picked cotton from the spindles and drops it into the collector.

Fig. 5: End-effector trajectories (A, B, and C) evaluated for harvesting a cotton boll cluster. Trajectory C achieved the best performance by approaching below the target cluster, moving upward through it, and retracting for doffing.



IV. Cotton Detection Model and Clustering Algorithm

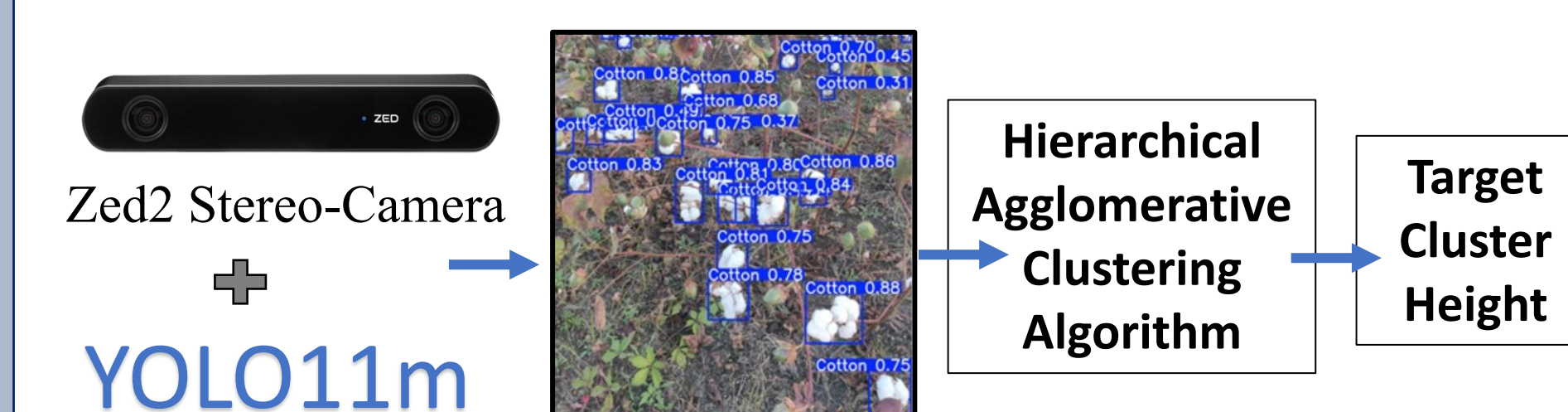


Fig. 6: Vision system's pipeline

V. Robotic Platform and Experimental Setup

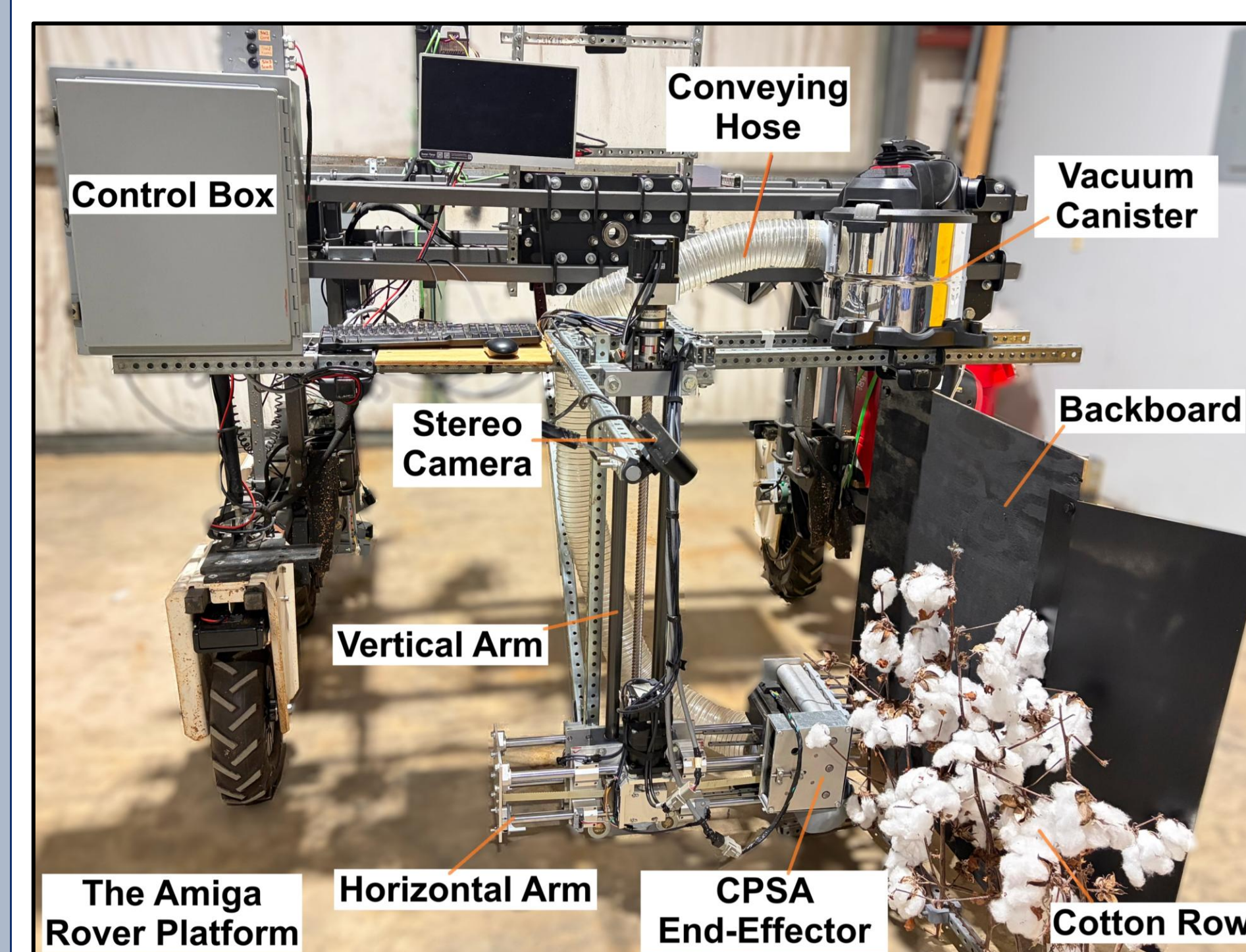
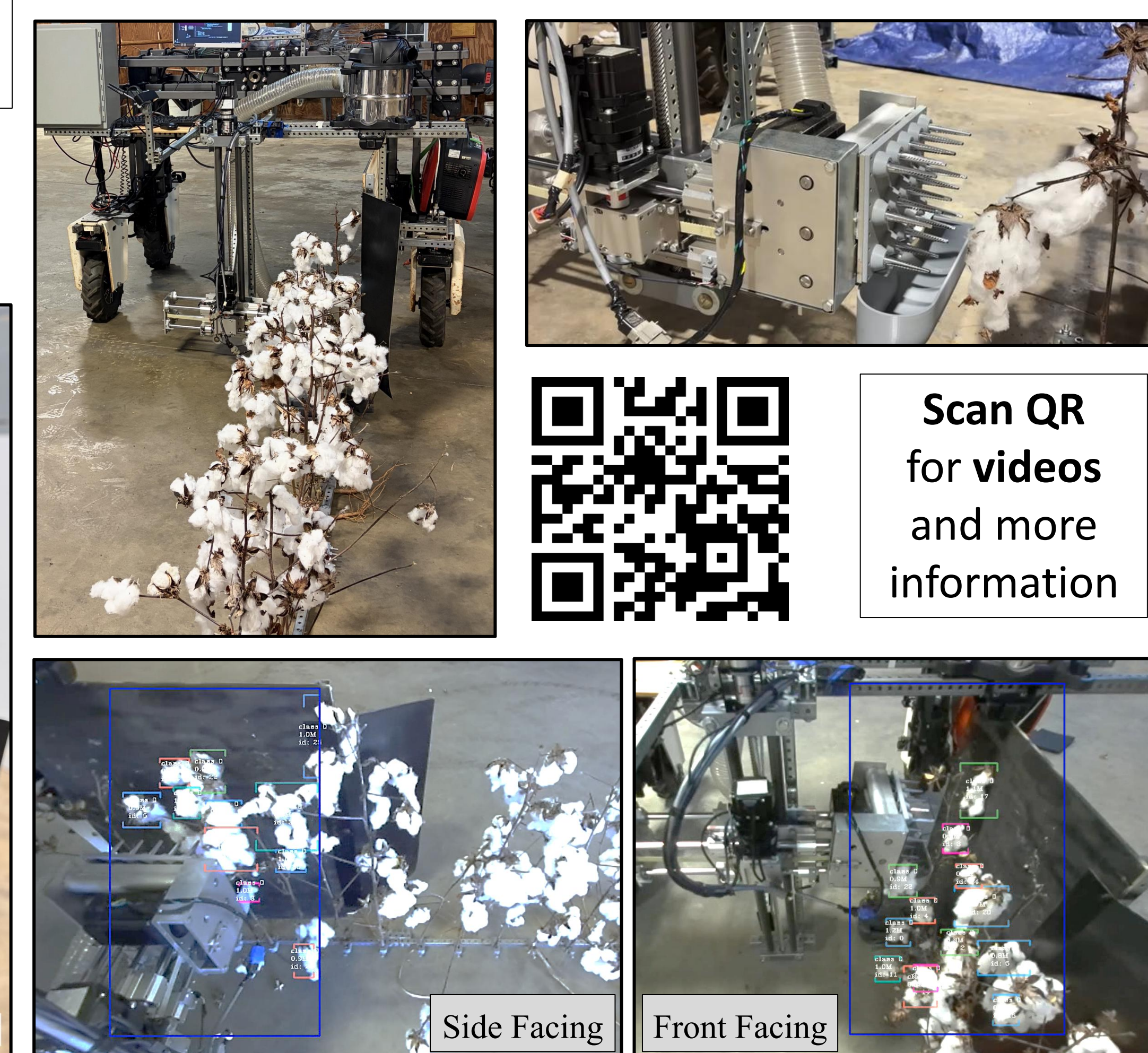
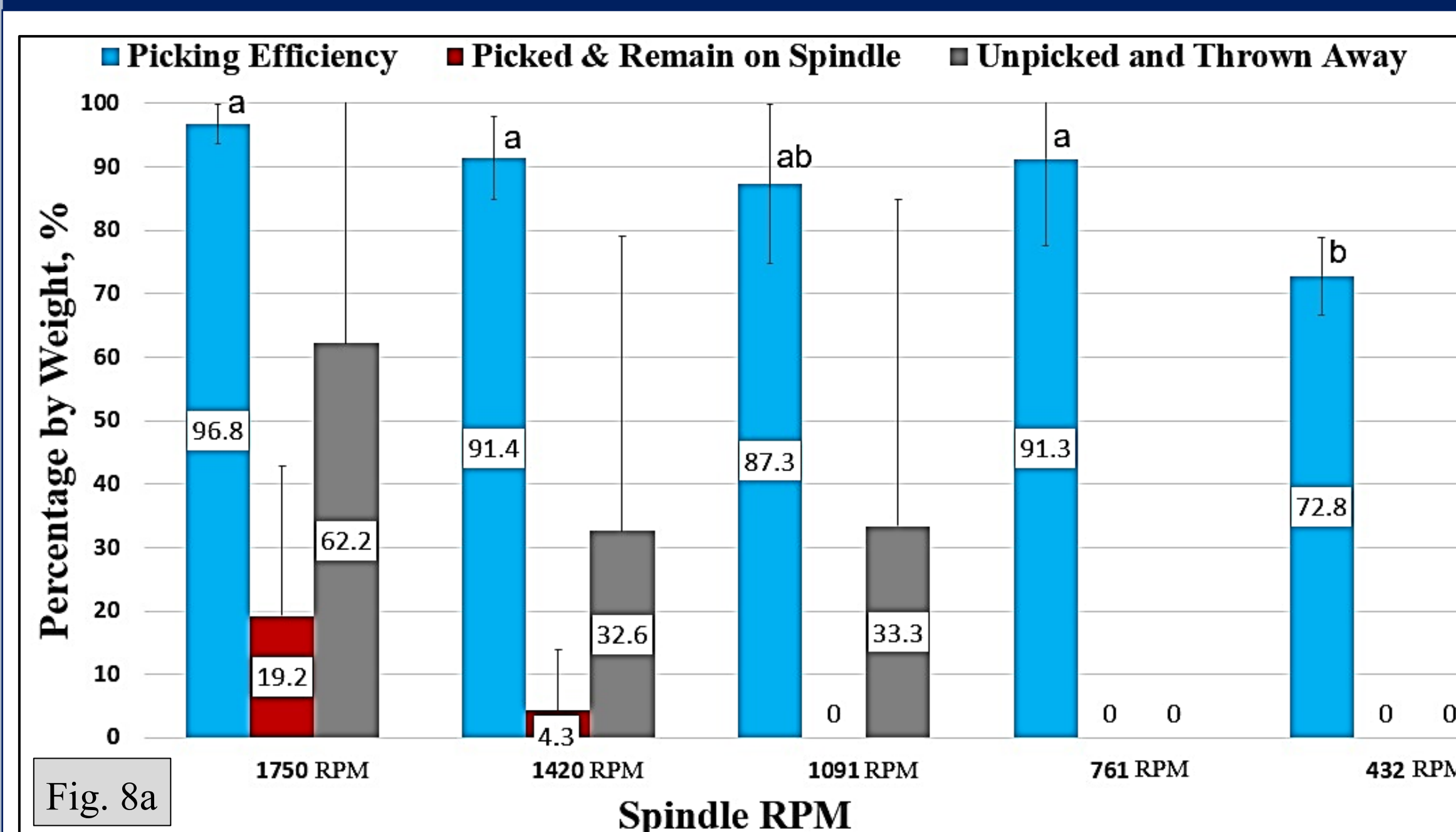


Fig. 7: Pictures showing the experimental setup and the robotic cotton harvesting system attached to the Amiga rover.

Model	mAP@0.5	mAP0.5:0.95	Precision	Recall	FPS
YOLO11m	0.933	0.791	0.930	0.859	35.52

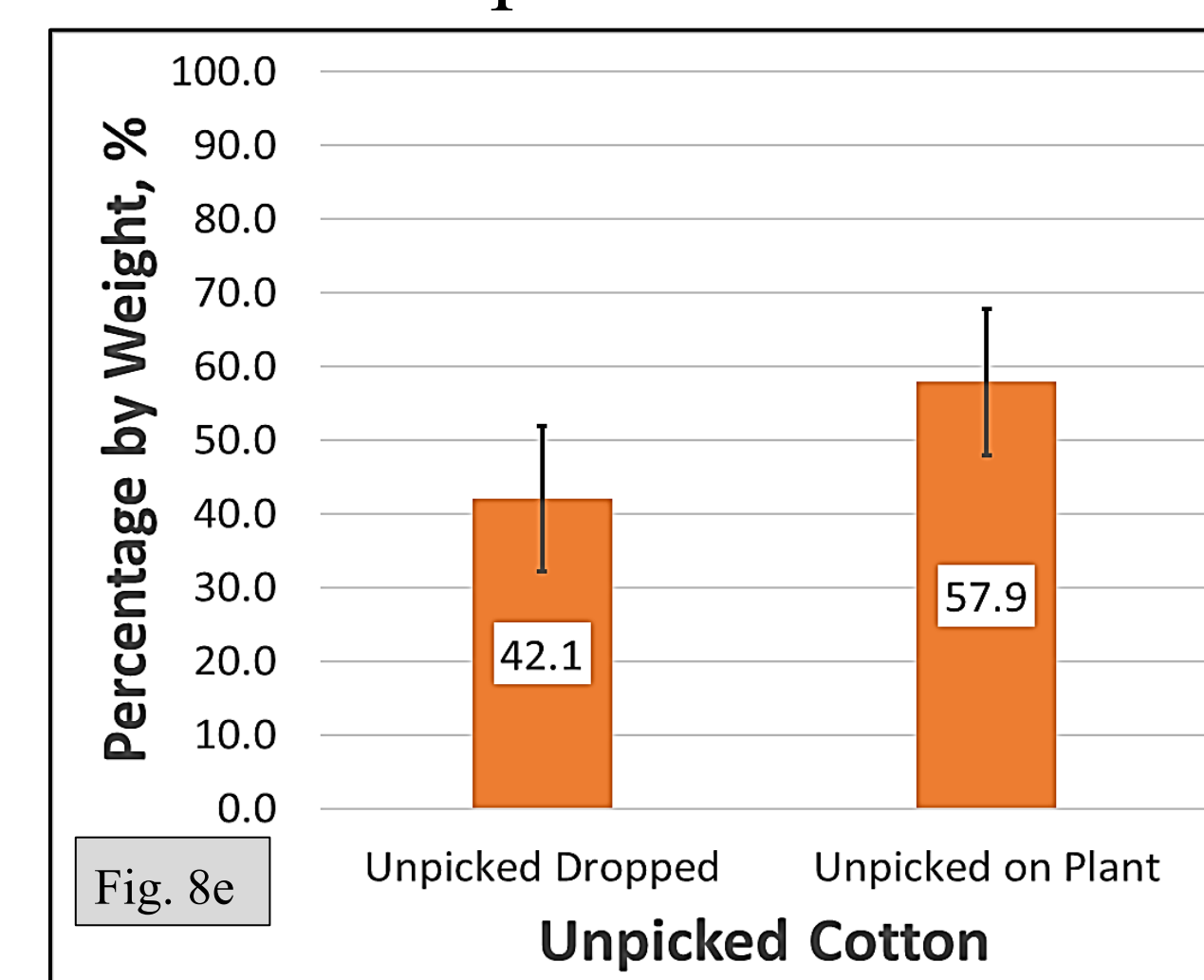
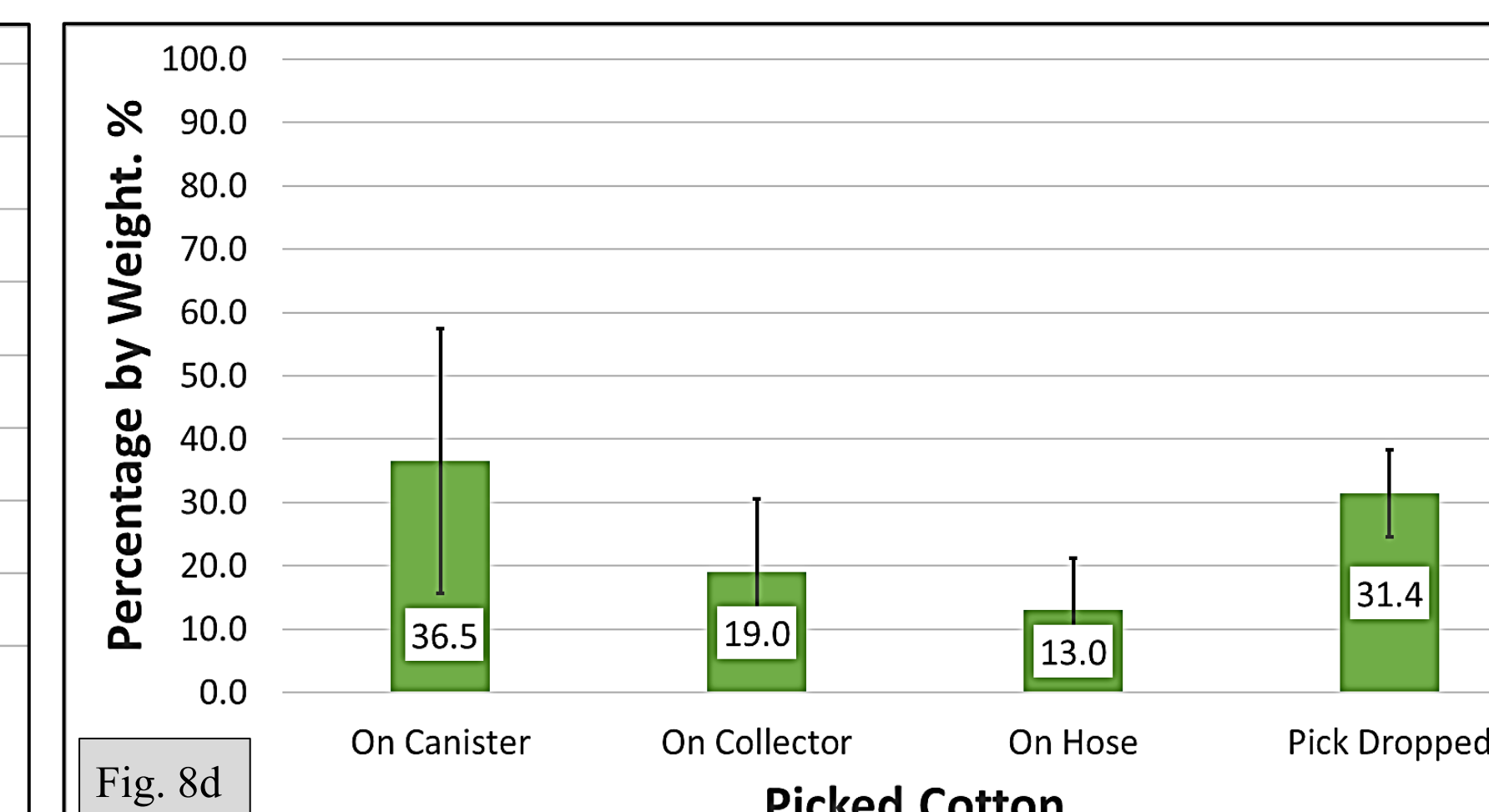
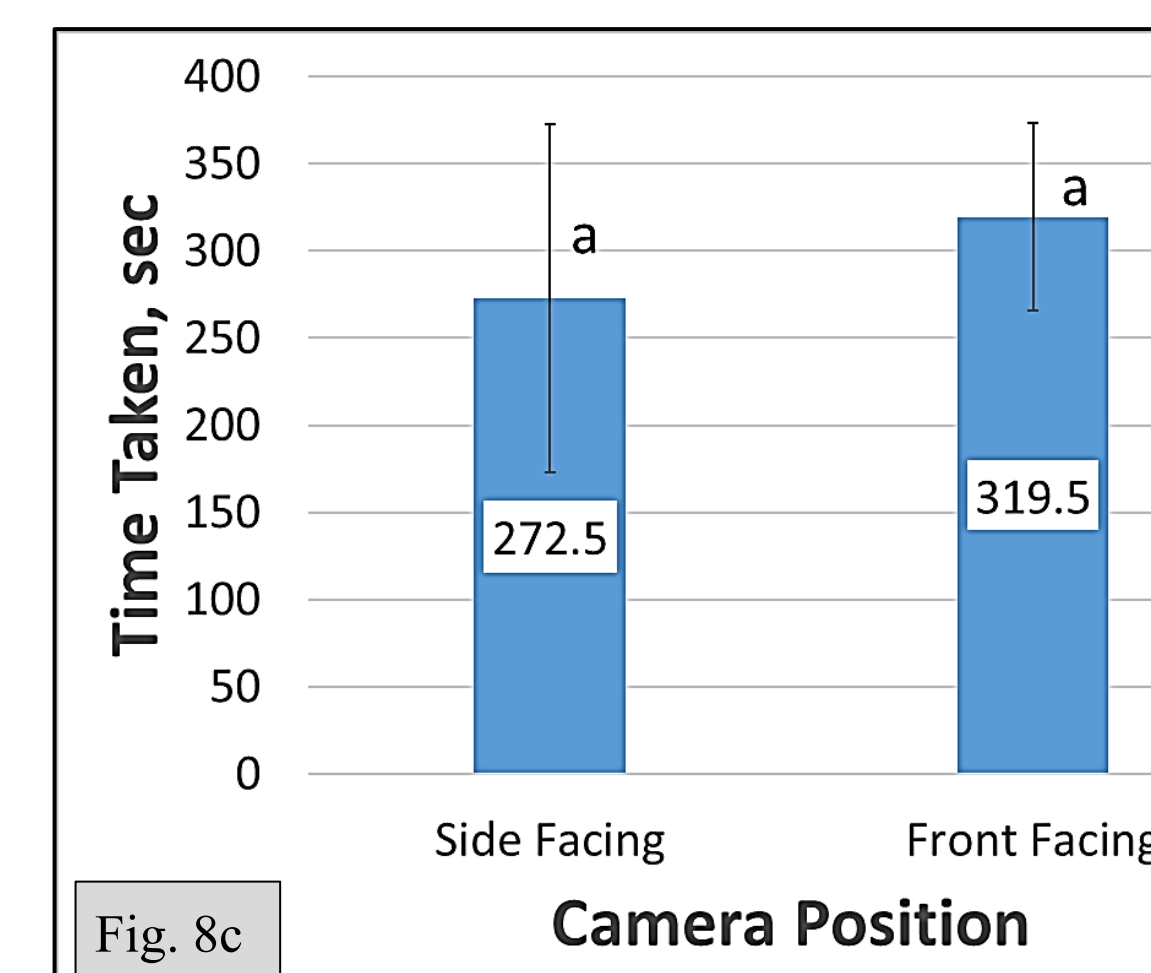
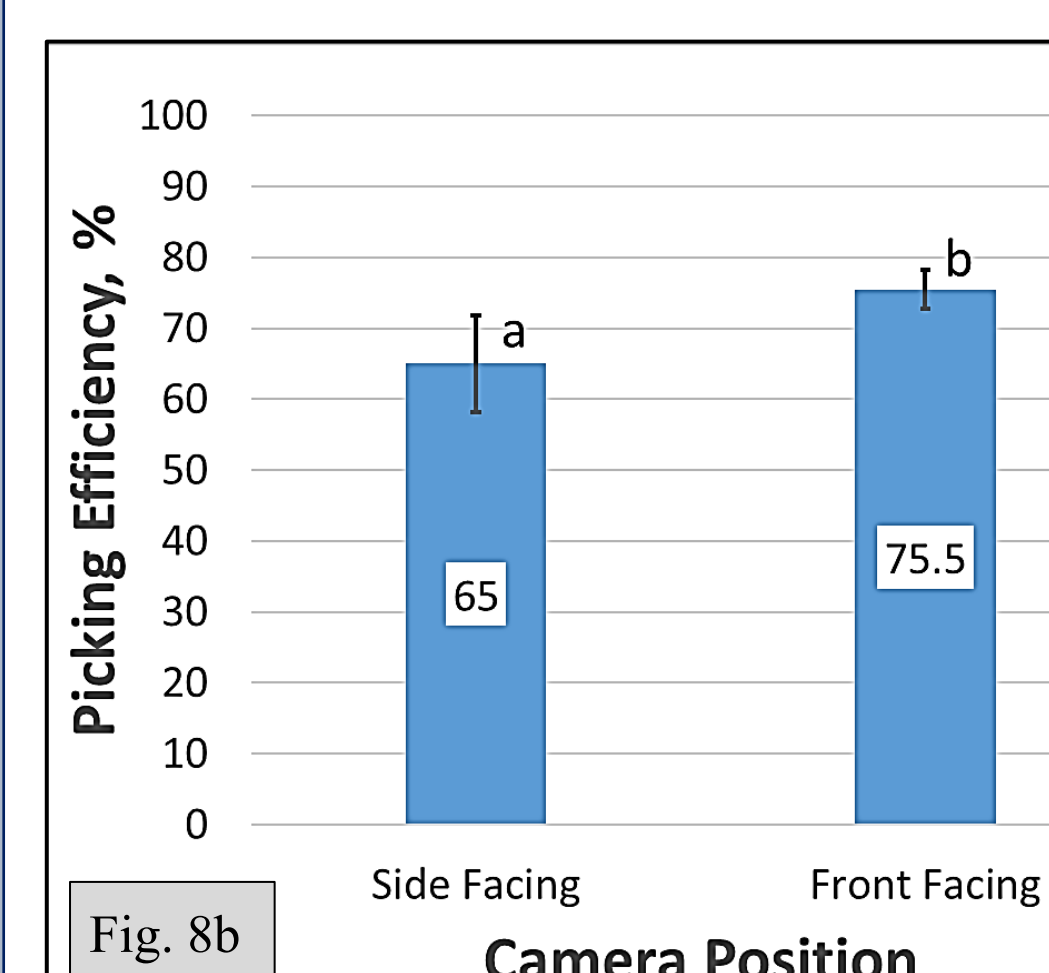


Results



II. Performances of Integrated System at Two Camera Positions

The system achieved an average picking efficiency of 75.5% with the front-facing camera, significantly higher than the side-facing position (Fig. 8b), and harvested a 6-ft row in ~5 min (~5.4 s/boll) (Fig. 8c). Harvested cotton accumulated in the canister, collector, and hose, while some fell to the ground (Fig. 8d). Of the unpicked cotton, ~42% were detached but dropped to the ground, and the rest remained unpicked on the plant (Fig. 8e).



Conclusions and Future Research

- A barbed spindle-array end-effector for a robotic system was designed and fabricated to harvest multi-bolls simultaneously, achieving optimal performance at arm trajectory C & 761 RPM spindle speed, reaching 75.5% picking efficiency at ~5.4 s/boll.
- Implementing continuous picking and doffing, and improving the vision and conveying systems, would significantly enhance system performance.
- The overall harvesting workflow with a baling mechanism will be investigated.

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Contact Info

